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ITESM Monterrey
B.S.E. Computer Science · GPA 4.0 · June 2020
Summa Cum Laude

EXPERIENCE

Microsoft · Windows Fundamentals

Senior Software Engineer · Lead, Cross-Vertical AI Evaluation *2021–Present*

AI EVALUATION FRAMEWORK

A production system where a frontier model analyzes Windows performance, power, and memory telemetry and proposes code optimizations engineers ship. I own the evaluation layer that decides which model suggestions are trustworthy.

- **Lifted production fix-acceptance from 54% to 78% on the frontier-model code-optimization pipeline by building its evaluation layer.** Currently the highest fix-acceptance rate among AI-assisted engineering workstreams in Windows, with only the early layers of the eval hierarchy in place. The framework evaluates whether model outputs are correct, grounded, actionable, and improving over time across Perf, Power, Memory, and future verticals on shared replay infrastructure.
- **Cut eval false positives ~35% and false negatives ~50% against a human-expert gold standard with a calibrated LLM-as-judge layer.** Scorers detect schema drift, parse failures, source-grounding errors, hallucinated APIs, unsupported mechanism and magnitude claims, classification inconsistencies, and overconfident recommendations. Judge calibration validated via Cohen's kappa.
- **Built reproducible replay infrastructure for model, prompt, advice-corpus, and context changes.** Each run holds inputs constant while varying one axis, then compares outputs across deterministic checks, cross-model disagreement, LLM-as-judge, human expert review, and production outcome tracking. Statistical methodology uses paired tests, bootstrap CIs, McNemar, Cohen's kappa, and Simpson's-paradox-aware aggregation, supporting model migration across Claude Opus (Anthropic SDK) and Azure-hosted GPT/o-series.
- **Defined the bi-directional taxonomy and feedback loop for AI-generated Windows optimizations.** Authored a taxonomy across algorithmic, caching, concurrency, I/O, memory, and anti-pattern classes; models use it for grounding, and novel model-proposed patterns route through validation gates before being admitted back into the corpus, creating a growing library of optimizations backed by shipped-fix evidence. Public post: Bi-Directional Class-of-Fix Taxonomies. Architected the end-to-end loop (telemetry-driven bug discovery → prompt assembly → model analysis → evaluation → code-change generation → automated PR → regression testing) that was adopted as the team's technical direction.

WINDOWS GLITCHINESS METRIC

- **Designed and shipped Glitchiness, a new metric for perceived Windows responsiveness, validated with human-subjects studies.** Delay-signal end-time clustering with span-based duration and long-delay promotion, deployed across the Windows fleet with daily failure-rate CoV 4.8%. A 41-participant controlled perceptual study plus an n=2,234 in-the-wild user-feedback study show users report sluggishness +9.3pp more often when Glitchiness flags a bad period (p=4.8e-5, Cohen's h=0.19). The methodology of operationalizing an inherently fuzzy human-judgment concept into a statistically-validated measurement transfers directly to evaluating subjective, hard-to-specify qualities of AI systems.
- **Connected memory pressure to perceived OS responsiveness at fleet scale.** Built an algorithm joining the Glitchiness device-day base to system memory telemetry, quantifying how memory pressure degrades perceived responsiveness with a clear dose-response in daily usage and reproducing the effect across CPU families and device populations. The finding is informing Windows memory-management strategy.

FLEET TELEMETRY & FORECASTING SYSTEMS

- **Led metric and forecasting systems for Windows performance decisions and rebuilt the underlying fleet telemetry infrastructure.** Created the LongInputDelays leadership metric with a normalized hits-per-million-devices framing that avoids population-shift artifacts. Built the SLT Glidepath forecasting system (least-squares regression on log-transformed census, two-phase hybrid growth model, R^2=0.90 backtested across active silicon). Rewrote PerfTrack targeting/routing after finding a measurement blind spot, growing retail performance telemetry from hundreds of thousands to millions of devices/day and multiplying the signal underneath every downstream Windows performance metric.
- **Earlier work (2021–2024).** Foundational telemetry-driven regression-detection platform that became the substrate for the current evaluation work; cross-org device-targeting pipeline scaled from 100K to 2.5M devices; executive metric system with distance-to-target attribution and predictive forecasting; population-shift detection separating product regressions from fleet-composition changes; introduction of mouse/keyboard latency as a first-class telemetry signal that Glitchiness was later built on.

Apple · Google

AI/ML & Software Engineering Intern *Summers 2017–2019*

- **Apple Siri Search (2019):** expanded Knowledge Card entity-type coverage and built a Hive/Impala automation pipeline for query-log-driven inline card refresh. **+214% entity coverage, +35% positive abandonment, +7% engagement** via A/B tests. **Google Cloud (2017):** multithreaded gRPC load generator and scheduler for next-generation network systems, **+31% network performance**. **Google Search (2018):** backend infrastructure for a food-ordering feedback system (Spanner schema migration, encrypted ingestion APIs, real-time partner-rating dashboards).

SKILLS

ML / AI Evaluation: LLM-as-judge calibration, scorer design, replay infrastructure, sycophancy / hallucination / scorer-gaming detection, frontier-model comparison (Claude Opus via Anthropic SDK, GPT-5, o-series), human-subjects studies, perceptual psychophysics.

Statistics: paired hypothesis testing, bootstrap CIs, McNemar, Cohen's kappa & h, Brier score, Simpson's-paradox-aware aggregation, calibration vs. human gold standard, time-series forecasting.

Systems: Python, C# / .NET 10 (Blazor Server, EF Core), TypeScript, SQL, Kusto/KQL, USQL/Scope (Azure Data Lake Analytics), Azure OpenAI, GitHub Copilot SDK, MCP, Azure Batch, Azure SQL.